NANYANG TECHNOLOGICAL UNIVERSITY

SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2 MH1301 Discrete Mathematics Tutorial 7

For the tutorial in Week 8 on 14 March, let us discuss the following questions.

Question 1. How many different possible simple graphs of n vertices and m edges are there? Assume that $G = (V, E)$ with $V = \{1, \ldots, n\}$.

Solution: For a graph with n vertices, there are $C(n, 2) = n(n-1)/2$ possible edges. There are $C(C(n, 2), m) = C(n(n-1)/2, m)$ possible ways to choose m edges out of the $n(n-1)/2$ possible edges. Therefore, there are $C(n(n-1)/2, m)$ different possible simple graphs with n vertices and m edges. For instance, for $n = 3$ and $m = 2$ there are 3 distinct simple graphs.

Question 2.

(a) Do simple, 3-regular graphs of 5 edges exist?
(b) How many graphs with 6 vertices and 14 edges (with vertex labels $\{1, \ldots, n\}$) are there?

Solution:

(a) Let n be the number of vertices in a simple graph. Due to the Handshaking Lemma, for 3-regular graphs of 5 edges, we would have $2 \cdot 5 = n \cdot 3$, so no such graphs exist, since n has to be an integer.
(b) A complete graph on 6 vertices has $C(6, 2) = 15$ edges, so we have to choose 1 edge for removal. Hence there are 15 such graphs.

Question 3.

(a) How many non-isomorphic simple graphs with 3 vertices and 2 edges are there?
(b) How many non-isomorphic simple graphs with 4 vertices and 2 edges are there?

Solution:

(a) For 3 vertices and 2 edges, the only possible shape of a graph is a path with 2 edges. The only difference between the graphs is which vertex is in the middle, 1, 2, or 3. Hence there are 3 simple graphs, but they are all isomorphic. So in this case there is only 1 graph up to isomorphisms, i.e., all graphs of 3 vertices and 2 edges are isomorphic.
(b) 2. The edges can have a vertex in common (this is a path with three vertices together with an isolated vertex), or not (two disjoint edges).

Question 4. (Ex. 10.2.18.) Show that in a simple graph with at least two vertices, there must be two vertices that have the same degree.

Solution: This question can be solved the same way we solve Ex. 6.2.32 (Tutorial 2) where we prove that at a party of at least two people, there are two people who know the same number of other people there. We just need to view people as vertices and two people knowing each other means there is an edge connecting the two vertices.

In a simple graph, any vertex has a degree between 0 to $n-1$.

Case 1: There is a vertex, call it A, that is adjacent to every other vertex in the graph. So any vertex has degree at least 1. Hence there are n vertices (pigeons) and the degree of any of the n vertices is between 1 and $n-1$ (total of $n-1$ pigeonholes). By the pigeonhole principle, there will be at least two vertices that have the same degree.

Case 2: There is no vertex that is adjacent to every other vertex in the graph. So there is no vertex with degree $n-1$ and so any vertex has degree between 0 to $n-2$. Again we have n vertices (pigeons) and $n-1$ choices for the degrees (pigeonholes) and so the conclusion holds for this case as well.

Question 5. How many non-isomorphic simple graphs with 4 vertices and 3 edges are there?

Solution: For a graph with 4 vertices (say v_1, v_2, v_3, v_4) and 3 edges, there are 3 non-isomorphic graphs:

- the path of length 3 (i.e. for $i = 1, 2, 3$, we have $\{v_i, v_{i+1}\}$ is an edge),
- a triangle (K_3) on 3 vertices, with one isolated vertex,
- a graph where one vertex is connected to the other 3.

These three graphs are clearly not isomorphic to each other (the graph with the triangle has an isolated vertex, the path does not, the third graph has a degree 3 vertex, which the others do not). There are no other graphs of 4 vertices and 3 edges that are not isomorphic to one of these. This is easily established by trying all possible graphs.
But more formally: any graph with degree 3 looks like the third graph. A graph of 4 vertices and 3 edges cannot have degree 1, because such a degree 1 graph can only have at most two edges. So there is a vertex of degree 2. Adding another edge in either completes the triangle or the path.

Question 6. (Ex. 10.2.60. and Ex. 10.2.65). The <u>complement</u> of a graph $G = (V, E)$ is the graph $\overline{G} = (V, \overline{E})$ where $\overline{E}$ contains exactly those edges that are <u>not</u> in E.

(a) Show that if G is a simple graph with n vertices, then the union of G and $\overline{G}$ is K_n.

(b) If G is a simple graph with 15 edges and $\overline{G}$ has 13 edges, how many vertices does G have?

Solution:

(a) By the definition "The <u>complement</u> of a graph $G = (V, E)$ is the graph $\overline{G} = (V, \overline{E})$, where $\overline{E}$ contains exactly those edges that are <u>not</u> in E", the union of the edges from G and $\overline{G}$ contains all possible edges in K_n.

(b) The two graphs G and $\overline{G}$ partition the edges of the complete graph, and the number of edges of a complete graph of n vertices is $C(n, 2) = 15 + 13 = 28$. This implies $n = 8$.

Question 7. Suppose that G and H are isomorphic simple graphs. Show that their complementary graphs $\overline{G}$ and $\overline{H}$ are also isomorphic.

Solution: Denote $G = (V_1, E_1)$ and $H = (V_2, E_2)$. Since G and H are isomorphic graphs, there exists a bijection $f : V_1 \to V_2$ and $\{u, v\} \in E_1$ with $u, v \in V_1$ if and only iff $\{f(u), f(v)\} \in E_2$, the contrapositive is also true, i.e.,

$$\{f(u), f(v)\} \notin E_2 \quad \text{if and only if} \quad \{u, v\} \notin E_1,$$

i.e., $\{f(u), f(v)\} \in \overline{E_2}$ if and only if $\{u, v\} \in \overline{E_1}$. Hence $\overline{G}$ and $\overline{H}$ are also isomorphic.

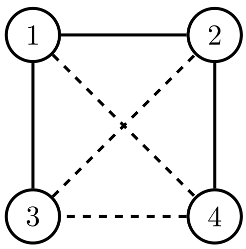
Question 8. A simple graph G is called self-complementary if G and $\overline{G}$ are isomorphic.

(a) Show that there are no self-complementary simple graphs with two nor three vertices.
(b) Find a self-complementary simple graph with four vertices.

Solution:

(a) There are no examples for $n = |V| \in \{2, 3\}$: if G is isomorphic to the complement of G, then both have the same number of edges. Furthermore, their union is K_n with $C(n, 2)$ edges, which is odd for both $n \in \{2, 3\}$.

(b) Here's an example for $n = 4$. By Part (a), the graph has $C(4, 2)/2 = 3$ edges.



The graph is a path 3–1–2–4. The complement is a path 1–4–3–2.
Hence an isomorphism is f with $f(3) = 1, f(1) = 4, f(2) = 3, f(4) = 2$.

Question 9.

(a) Find a self-complementary simple graph with five vertices.
(b) For which integers n is C_n self-complementary ?

Solution:

(a) Let $G = (V, E)$ with

$$V = \{a, b, c, d, e\} \qquad \text{and} \qquad E = \{\{a, b\}, \{b, c\}, \{c, d\}, \{d, e\}, \{e, a\}\}.$$

Then

$$\overline{G} = \{V, \overline{E}\} = \{\{a, c\}, \{c, e\}, \{e, b\}, \{b, d\}, \{d, a\}\}$$

with the bijective mapping $f(a) = a, f(b) = c, f(c) = e, f(d) = b, f(e) = d$.

(b) The number of edges of C_n is n, hence the self-complementary contains n edges as well. Note that the union of the graph and its complementary form K_n, hence $n + n = C(n, 2)$, which implies $n = 5$.

Question 10. How many subgraphs with at least one vertex does K_3 have?

Solution: We enumerate the number of vertices of this subgraph:

Case 1: The subgraph has only one vertex. Therefore the subgraph contains 0 edges. There are 3 possibilities in this case.

Case 2: The subgraph has two vertices. There are $C(3, 2) = 3$ ways to choose the two vertices. Then there could be an edge between them or no edge. So in total there are $3 \times 2 = 6$ possibilities.

Case 3: The subgraph has all three vertices. There are three possible edges, each may or may not be present in the subgraph. So there are in total $2^3 = 8$ possibilities.

Summing all cases together, there are in total $3 + 6 + 8 = 17$ possible subgraphs.

Answer.

1) $C(n(n-1)/2, m)$.
2(b) 15.
3(a) 1; 3(b) 2.
5) 3. Explain why they are not isomorphic.
6(b) 8.
9(b) 5.
10) 17.